

An experimental investigation of conjunct agreement in Turkish

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(Partial) conjunct agreement

- With conjoined phrases, one of the conjoined elements or more can be the target of agreement:

1) Raḥḥ tuktub/yuktub-u [il-mudiiraat w musaaʿiddiin]=hum taqriir.
FUT write.3F.SG/3M.PL [the.managers.F and assistants.M]3PL=their.report
‘The (female) managers and their (male) assistants will write a report.’

first conjunct (FCA) full (FA), aka. resolved agreement
Jordanian Arabic
(Al Khalaf., 2021)

2) Maria chce, żebym [ja i mój sąsiad] wyszedł.
M. wants that.COND.1SG [I and my neighbor.M.SG]1PL left.M.SG
‘Maria wants me and my neighbor to leave.’

last conjunct (LCA)

Polish
(Citko, 2018)

3) [Yıldız ve ben] geçenlerde müdürle konuştum/konuştuk.
[Y. and I]1PL recently with.principal talked.1SG/1PL
‘Yıldız and I talked with the principal recently.’

Turkish
(Köylü, 2018;
Paparounas & Akkuş, 2023);
Tat & Kornfilt, 2022)

Accounts for partial conjunct agreement

Why do we get partial agreement?

ConjP is **ϕ -deficient**;

no ordinary route for agreement, leading to different ‘strategies’.

How do we account for it?

Syntactic accounts: derived through hierarchical structure (e.g., equidistance of Spec of ConjP and first conjunct, movement, order of operations)

(Bošković, 2009; Murphy & Puškar, 2017; van Koppen, 2005, 2006)

Accounts with post-syntactic processes:

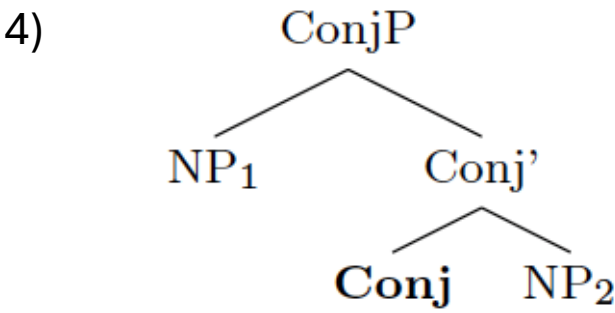
1. AGREE-LINK: establishing a relation between a syntactic head and an object
2. AGREE-COPY: transferring ϕ features (of a single conjunct or ConjP) to the head
→ agreeing with **linearly closer** noun post-syntactically, component after linearization

(Benmamoun et al., 2009; Bhatt & Walkow, 2013; Marušič et al., 2015)

*How acceptable are **partial agreement** patterns in different (syntactic) contexts in Turkish?*

TODAY: We show evidence for partial agreement that varies **across different levels**

Preliminaries for Turkish



(Johannessen, 1998)

5) [Her_i kadın ve pro_i çocuğ-u] bu durumdan memnun.
 [every woman and kid-POSS] this situation be.satisfied.with
 ‘Every woman_i and her_i kid are satisfied with this situation.’

Paradigm 1: verbal predicates ending with the past tense marker *–DI* or the conditional marker *–sA*

Paradigm 2: elsewhere

	Paradigm 1		Paradigm 2	
	SINGULAR	PLURAL	SINGULAR	PLURAL
1ST	-m	-k	-(y)Im	-(y)Iz
2ND	-n	-nIz	-sIn	-sInIz
3RD	-Ø	-Ø / -lAr	-Ø	-Ø / -lAr

(For accounts with three agreement paradigms, see Güneş, 2020, 2021; Neu, 2024)

Methodology in Expt1-3.

Yıldız ve ben geçenlerde müdürle konuştum.

Kesinlikle duyabileceğim/söyleyeceğim bir cümle değil ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 *Kesinlikle duyabileceğim/söyleyeceğim bir cümle*

*Definitely not a sentence that I might
produce or hear from Turkish speakers*

*Definitely a sentence that I might
produce or hear from Turkish speakers*

→ aiming to tap into acceptability of sentences in **daily speech**

- Relatively large sample sizes
- Fillers: 24 grammatical, 24 ungrammatical, and 24 with coordinated objects
- Bayesian, *maximal*, mixed-effects *ordinal* regression models

Expt1: Is partial agreement due to coordination of clauses?

Partial conjunct agreement may be due to coordination of units larger than NPs, e.g., clauses, followed by Right Node Raising or Ellipsis.

- 6) [Yıldız ve ben] geçenlerde müdürle konuş-tu-**m**.
[Y. and **I**] recently principal.COM talk.PST-**1SG**
'Yıldız and I talked with the principal recently.'

[[_{IP} Yıldız ~~talked with the principal~~] and [_{IP} I talked with the principal]]

(Aoun et al., 1994; Hudson, 1970)

? Is partial agreement observed when clausal coordination is not possible?

→ testing with **collective** verbs

Expt1: Is partial agreement due to coordination of clauses?

2x3 design, crossing VERB TYPE and AGREEMENT (**full (FA)**, **last conjunct (LCA)**, **first conjunct (FCA)**)

7) a. Distributive verbs:

[**Yıldız ve ben**] ofisteki görüşme için telaşlan-dı-**k/m/∅**.

[Y. and I] office.at meeting for worry-PST-1PL/1SG/3SG

b. Collective verbs:

[**Yıldız ve ben**] ofisteki görüşme için toplan-dı-**k/m/∅**.

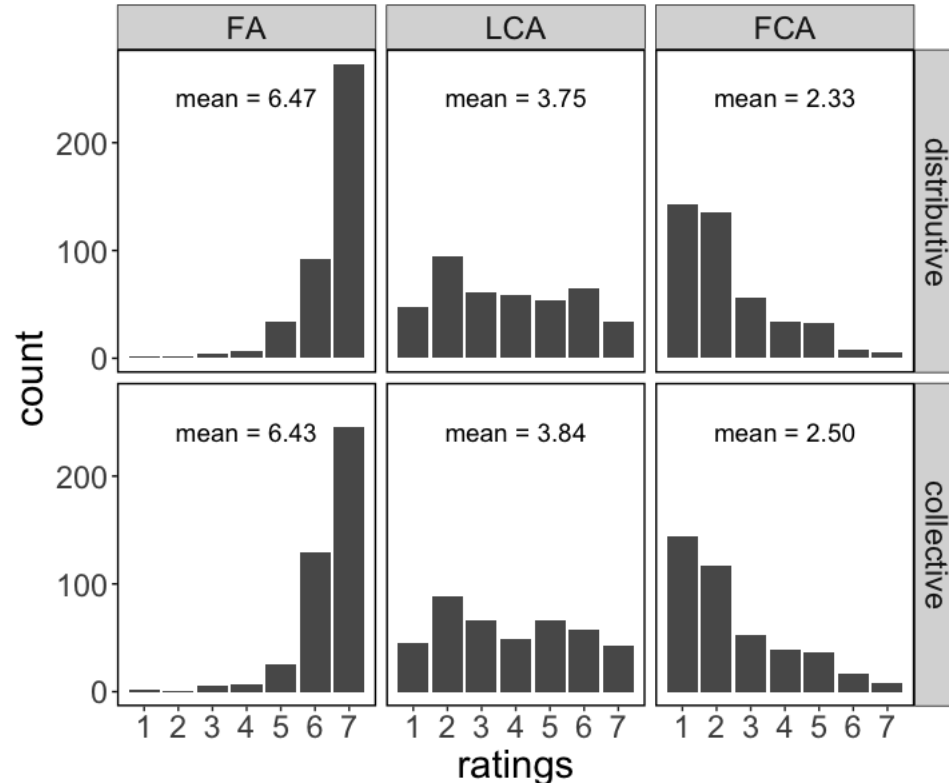
[Y. and I] office.at meeting for gather-PST-1PL/1SG/3SG

‘Yıldız and I {worried about/gathered for} the meeting at the office.’

($N_{participant}=69$, $N_{item}=36$)

Expt1: Population-level

Y. and I ... Verb



FA > LCA > FCA

with both distributive and collective verbs
with more slightly pronounced diffs in distributive

Predicate : RAvsLCA/FCA: Est. 0.27, 95% CI: [0.01, 0.53]

Predicate : LCAvsFCA: Est. 0.04, 95% CI: [-0.21, 0.29]

RA vs. LCA/FCA (distributive): Est. 3.52, 95% CI: [3.17, 3.89]

LCA vs. FCA (distributive): Est. 1.41, 95% CI: [1.09, 1.75]

RA vs. LCA/FCA (collective): Est. 3.28, 95% CI: [2.93, 3.63]

LCA vs. FCA (collective): Est. 1.40, 95% CI: [1.03, 1.77]

FCA ~ Ungrammatical ($M = 2.41$, $M = 2.43$)

→ partial agreement is not due to clausal coordination only

- similar experimental findings from

Slovenian, Bosnian/Croatian/Serbian (Arsenijevic et al., 2017)

English (Clifton & Frazier, 2012)

→ **Linearity** effects in ConjP

Expt2: Agreement with disjunctive ‘veya’?

DisjP is also **ϕ -deficient**, BUT DisjP doesn't denote semantic plurality

(Peterson, 2004)

- DisjP with two *singular* NPs in rating tasks in English or Italian:

(Foppolo & Staub, 2020)

8) The lawyer or the accountant **is/are** ...

overall degraded ratings, with no clear preference for singular or plural agreement

→ **Grammatical gap**: non-deterministic, variable decisions about verb number with DisjP

- DisjP with *number-mismatching* NPs in production tasks:

9) a. **Are** the horses or the clock red?

b. Can you tell me if the horses or the clock **is** red?

(Haskell & MacDonald, 2005)

→ Tendency to agree with linearly closer NP → linearity effects

? What agreement strategies are available in Turkish?

Expt2: Agreement with disjunctive ‘veya’?

2x2x3 design, crossing CONJUNCTION TYPE (‘and’/‘or’), ~~PREDICATE TYPE (verbal/nonverbal)~~ and AGREEMENT (**full (FA)**, **last conjunct (LCA)**, **first conjunct (FCA)**)

10) a. Coordination:

[**Yıldız ve ben**] saatler öncesinden var-ır/müsait-**ız/ım/ø**.

[Y. and I] hours in.advance arrive-AOR/be.available-PST-1PL/1SG/3SG

b. Disjunction:

[**Yıldız veya ben**] saatler öncesinden var-ır/müsait-**ız/ım/ø**.

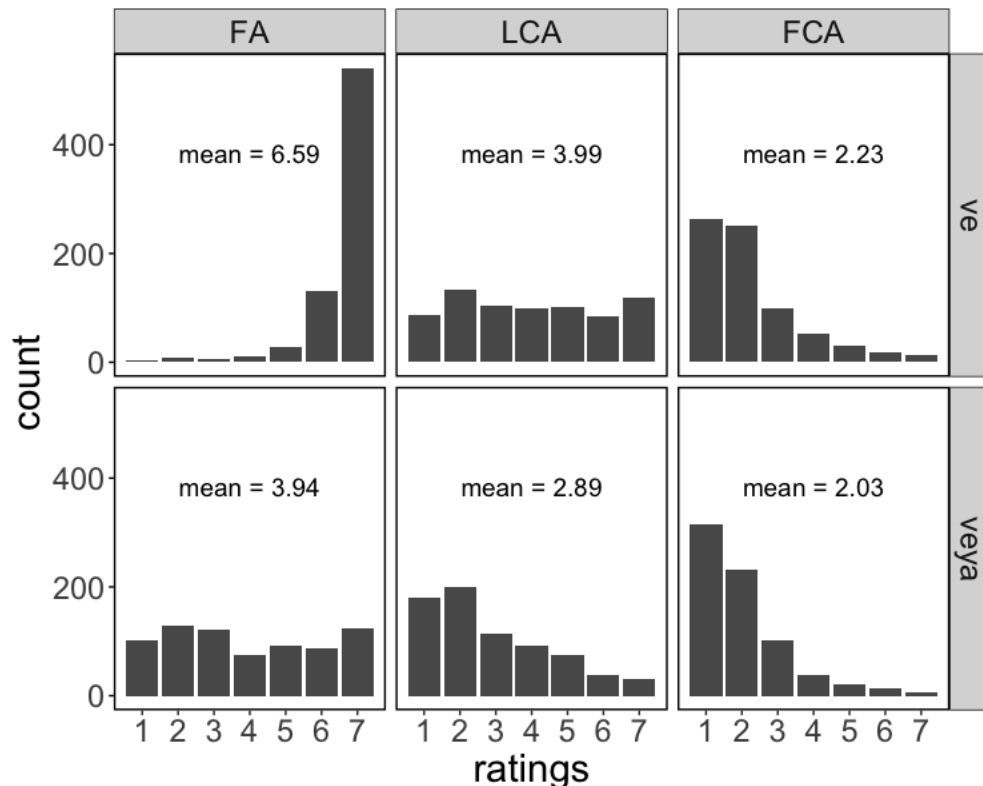
[Y. or I] hours in.advance arrive-aor/be.available-1pl/1sg/3sg

‘Yıldız {and/or} I will {arrive/be available} hours in advance.’

($N_{participant}=93$, $N_{item}=48$)

Expt2: Population-level

Y. and/or I ... Verb



Overall **lower** ratings with DisjP

Est. 1.24, 95% CI: [1.06, 1.42]

→ **grammatical gap** in verb number with DisjP

(Foppolo & Staub, 2020; Keung & Staub, 2018)

FA > **LCA** > **FCA**

in both coordination and disjunction

with more pronounced differences in coordination

Conjunction*RAvsLCA/FCA: Est. 2.08, 95% CI: [1.77, 2.42]

Conjunction*LCAvsFCA: Est. 0.64, 95% CI: [0.40, 0.88]

RA vs. LCA/FCA (and): Est. 2.64, 95% CI: [2.37, 2.90]

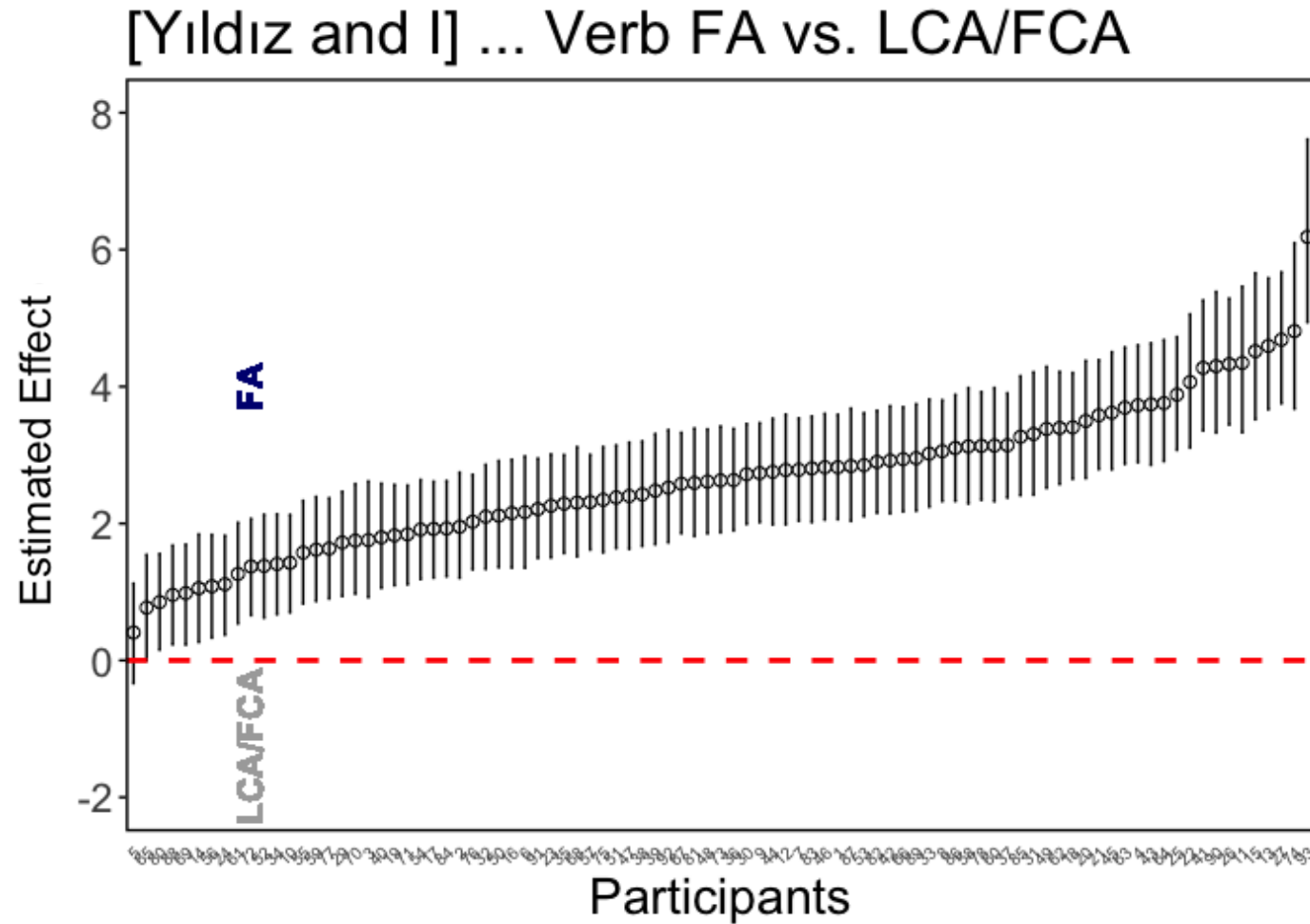
LCA vs. FCA (and): Est. 1.54, 95% CI: [1.25, 1.83]

RA vs. LCA/FCA (or): Est. 1.05, 95% CI: [0.86, 1.24]

LCA vs. FCA (or): Est. .069, 95% CI: [0.54, 0.84]

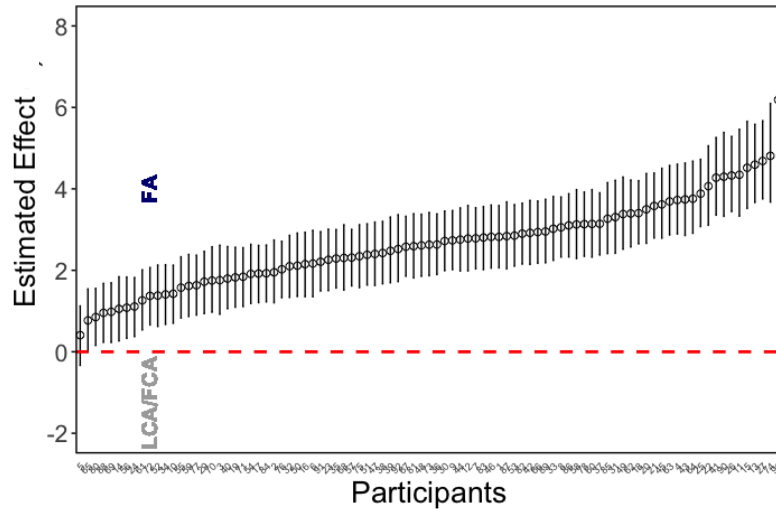
→ **linearity** effects in DisjP as well

Expt2: Speaker-level

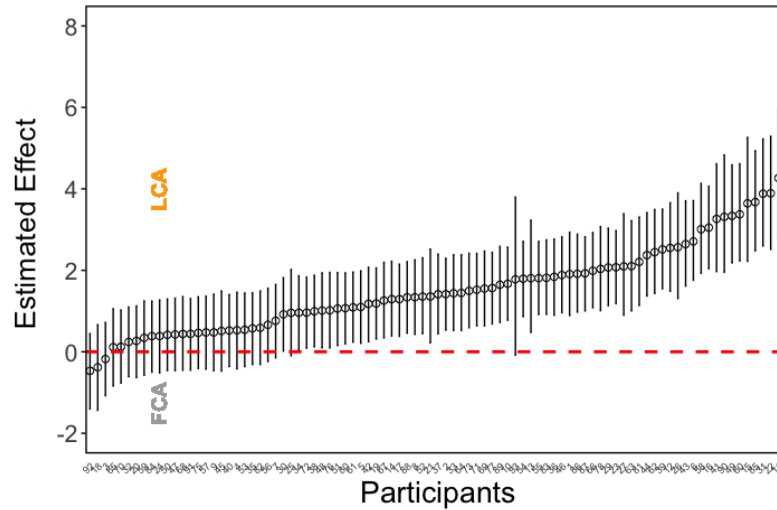


Expt2: Speaker-level

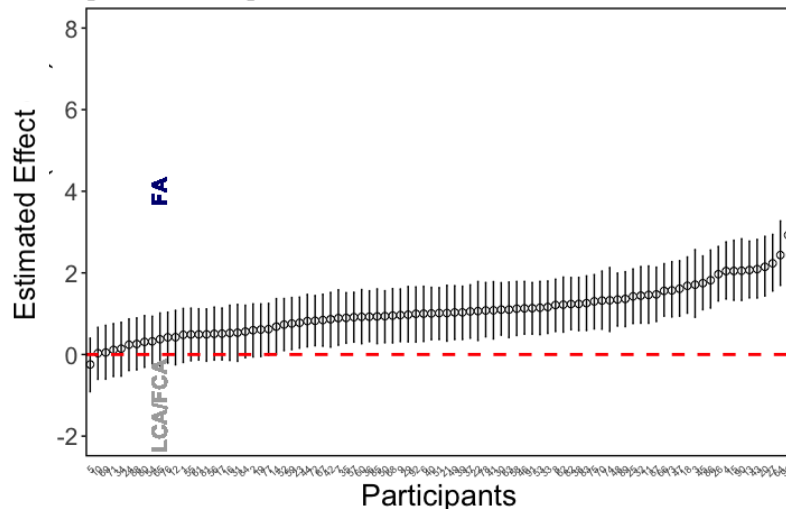
[Yıldız and I] ... Verb FA vs. LCA/FCA



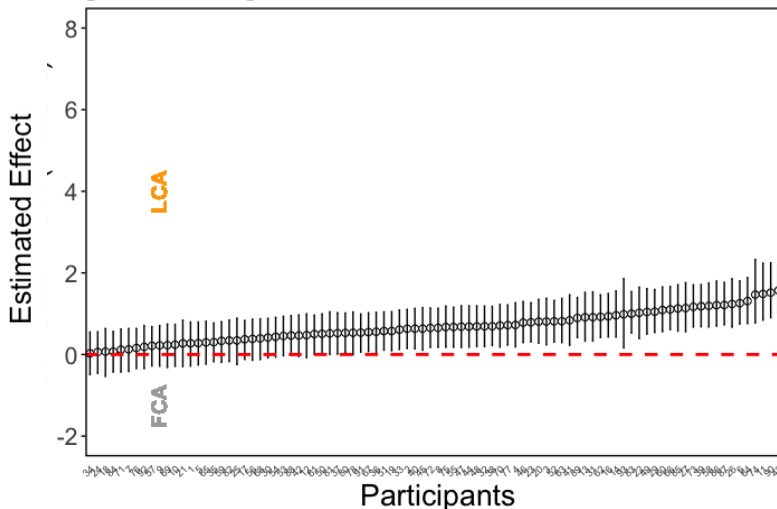
[Yıldız and I] ... Verb LCA vs. FCA



[Yıldız or I] ... Verb FA vs. LCA/FCA



[Yıldız or I] ... Verb LCA vs. FCA



Conjunction:

FA > LCA > FCA

between-speaker variance

ask me about within-speaker correlations in Q&A!

Disjunction:

FA > LCA > FCA

less between-speaker variance

Interim Summary

So far, for [**Y. and I ... Verb**];

FA is preferred,

LCA is better than **FCA**

This could be because:

? 'I' is linearly closer to the verb

[&P NP1 and **NP2**] Verb



Verb [&P **NP1** and NP2]



(Köylü, 2018)

? 'I' is 1sg.pronoun (higher in person hierarchy)

1/2 > definite specific NPs

(Ince, Aygen & Aydın, 2015)

Expt3: Is partial agreement dependent on conjunct position and/or conjunct types?

2x2x3 design, crossing CONJUNCT TYPE (1+2SG.pronoun / 1SG.pronoun+proper name),

CONJUNCT ORDER (preverbal/postverbal) and AGREEMENT (**full (FA)**, **last conjunct (LCA)**, **first conjunct (FCA)**)

11) a. Preverbal:

[**Ben ve** Yıldız] ... var-ir-**iz**/∅/ım.

[I and Y.] arrive-AOR-1PL/3SG/1SG

Does 'I' lead to FCA?

[**Ben ve** sen] ... var-ir-**iz**/**sin**/ım.

[I and you] arrive-AOR-1PL/2SG/1SG

Does 'you' lead to LCA?

b. Postverbal:

Var-ir-**iz**/**im**/∅ ... [**Yıldız ve** ben].

arrive-AOR-1PL/1SG/3SG [Y. and I]

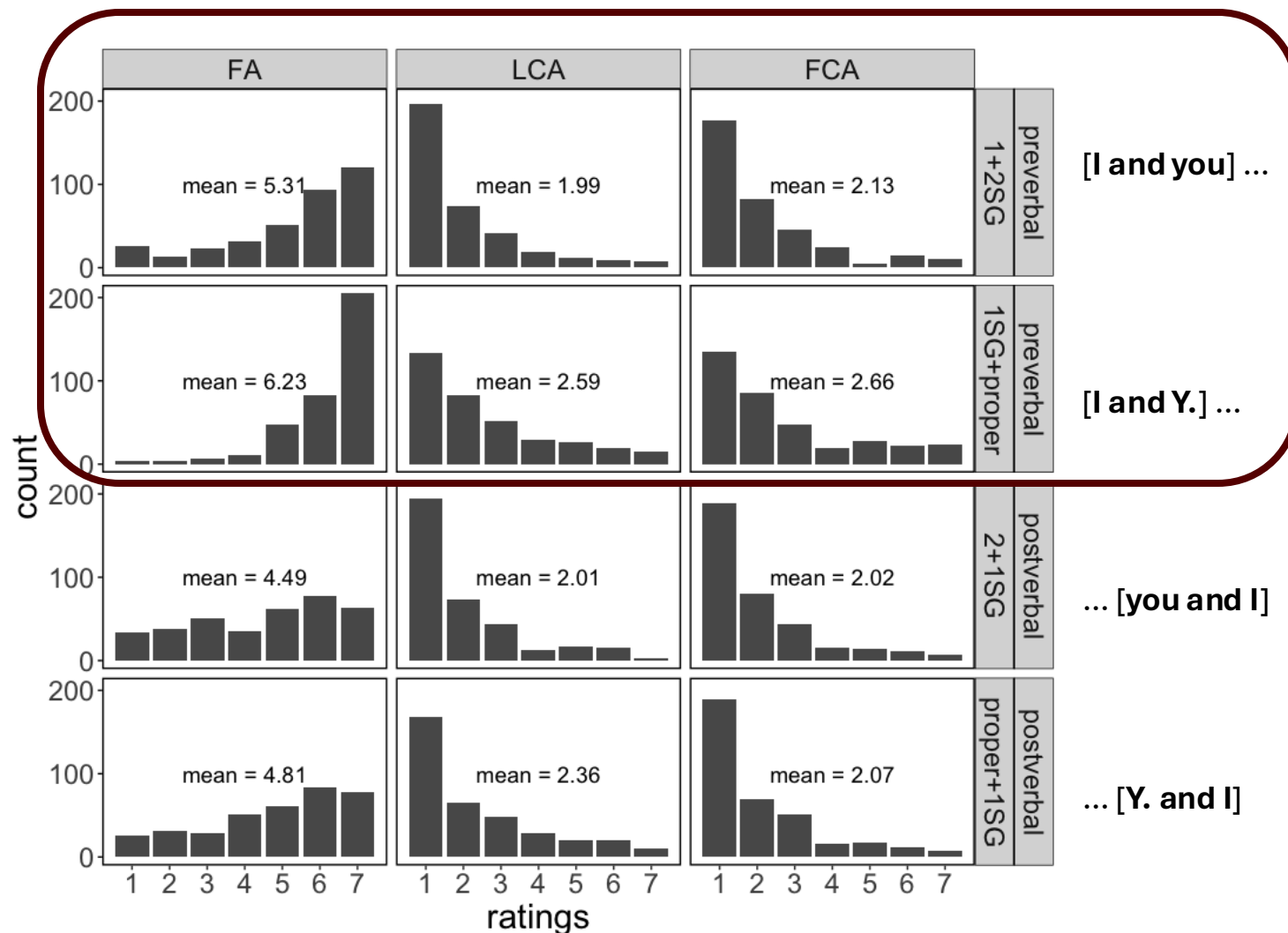
Is LCA available with 'I' in postverbal?

Var-ir-**iz**/**sin**/ım ... [**sen ve** ben]

arrive-AOR-1PL/2SG/1SG [you and I]

'I and {Yıldız/you} will arrive...'

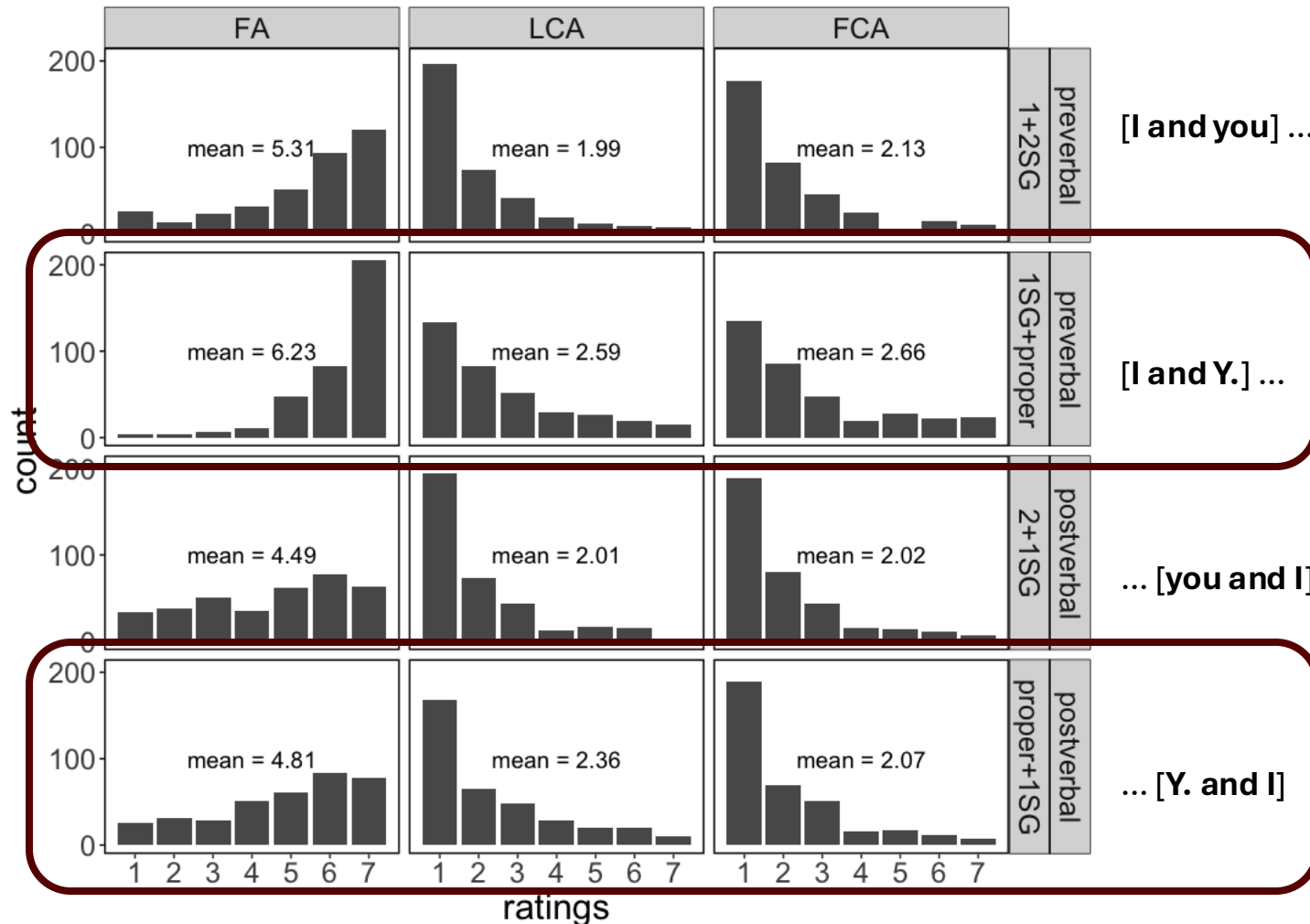
Expt3: Population-level



Preverbal subjects: higher ratings

Est. 0.59, 95% CI: [0.44, 0.75]

Expt3: Population-level



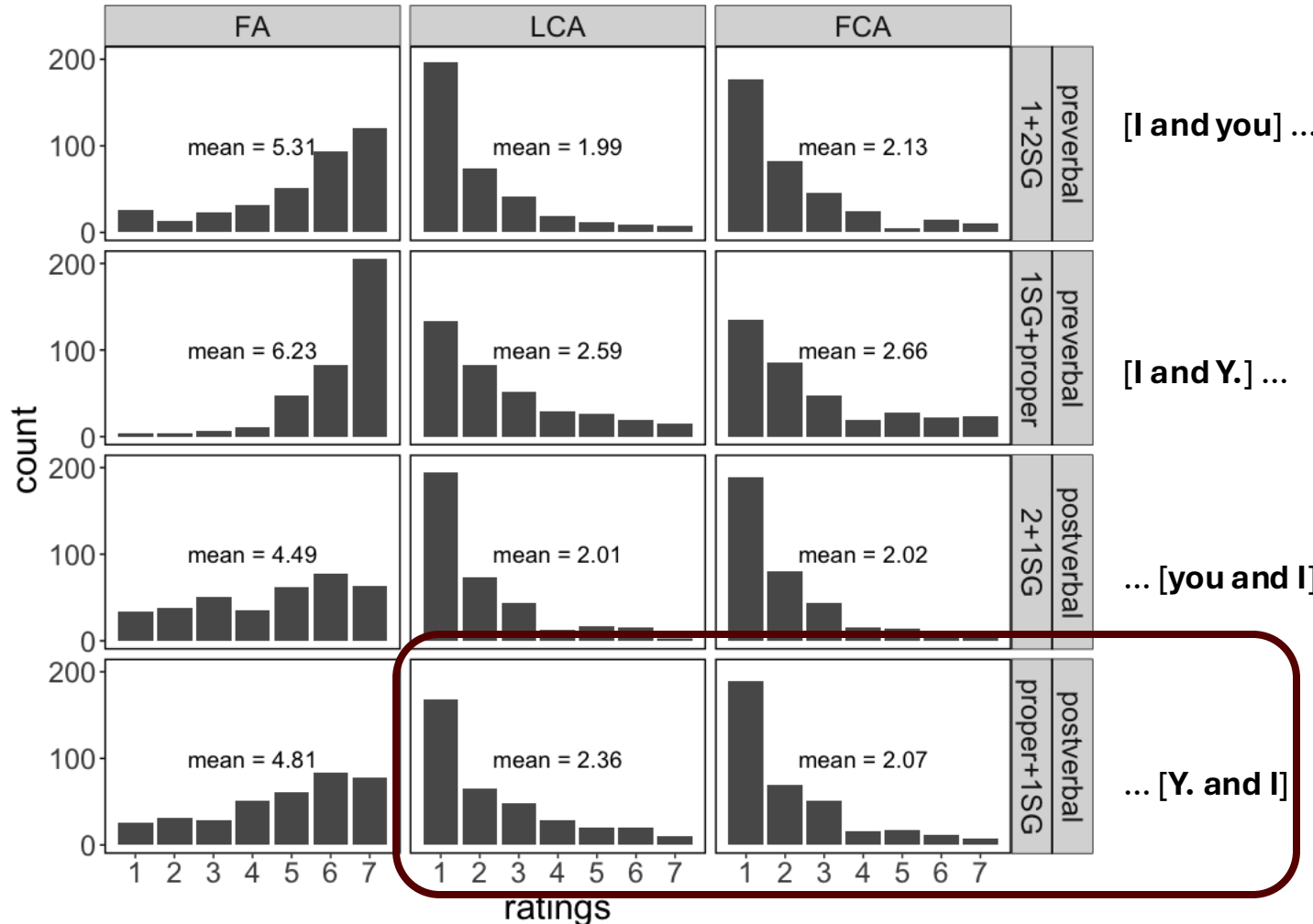
Preverbal subjects: higher ratings

Est. 0.59, 95% CI: [0.44, 0.75]

Pronoun + proper name: higher ratings

Est. -0.54, 95% CI: [-0.64, -0.43]

Expt3: Population-level



[I and you] ...

Preverbal subjects: higher ratings

Est. 0.59, 95% CI: [0.44, 0.75]

[I and Y.] ...

Pronoun + proper name: higher ratings

Est. -0.54, 95% CI: [-0.64, -0.43]

... [you and I]

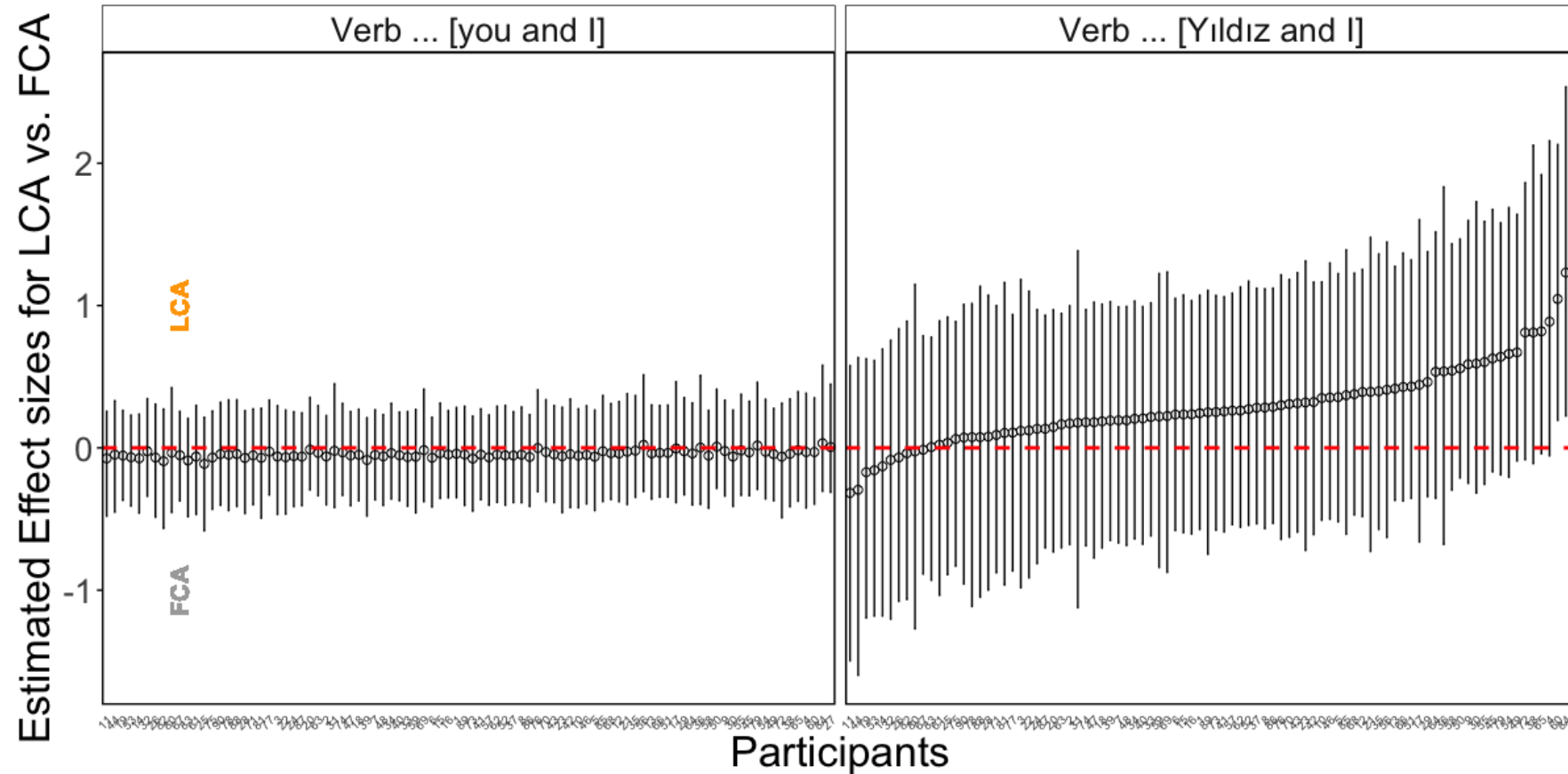
Some evidence for agreement with **linearly distant 'I'**:

Verb ... [&_PY. and I]

LCA > FCA

Est. 0.29, 95% CI: [0.07, 0.53]

Expt3: Speaker-level



tighter distribution around **zero** (no evidence for an effect)
with little *variance* between- and within-speakers

Overall preference for **LCA**
with *variance* between- and within-speakers

Discussion

- Partial conjunct agreement in Turkish is not due to clausal oordination
 - Comparable ratings with **distributive & collective** verbs
- Degraded ratings for all three agreement patterns with **DisjP**
 - **Grammatical gap**
- Evidence for **LCA** with **1sg.pronoun** (combined with a proper name);
 - When it is linearly closer or distant
 - Degraded acceptability with post-verbal subjects
 - **Variance** across and within speakers with **LCA**

(Arsenijevic et al., 2017; Clifton & Frazier, 2012)

(Foppolo & Staub, 2020; Keung & Staub, 2018)



- Future experiments to reveal effects of linearity and **person hierarchy**:

(Günay & Atlamaz, 2024)



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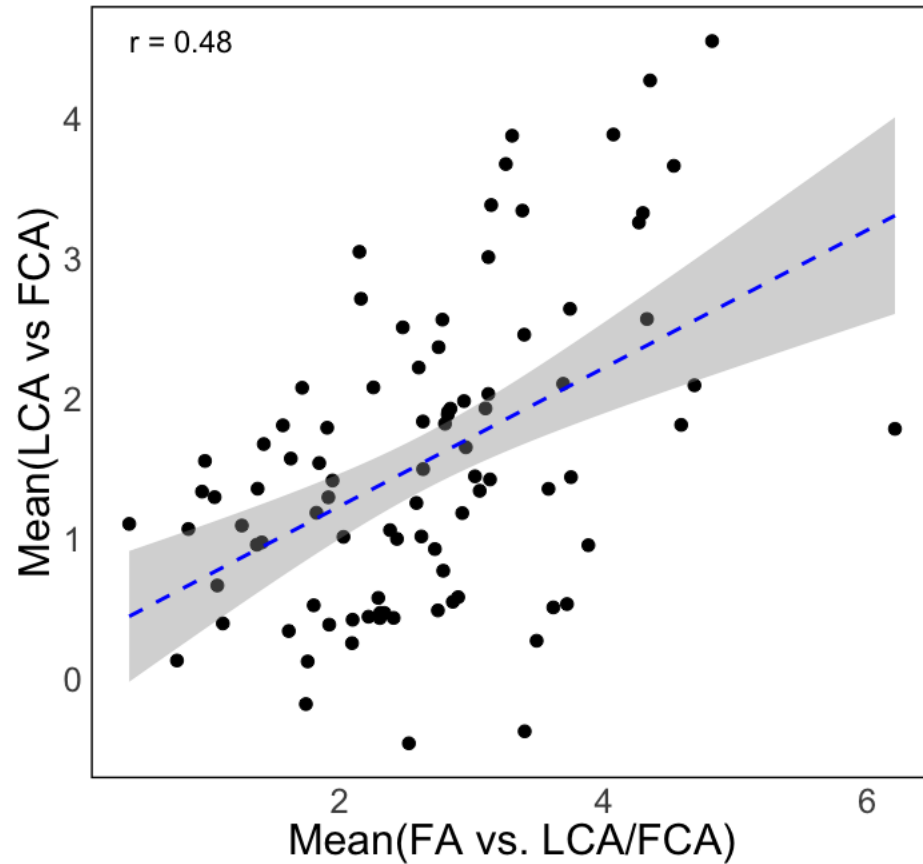
the Psycholinguistics and Syntax Workshops at UMass Amherst

References

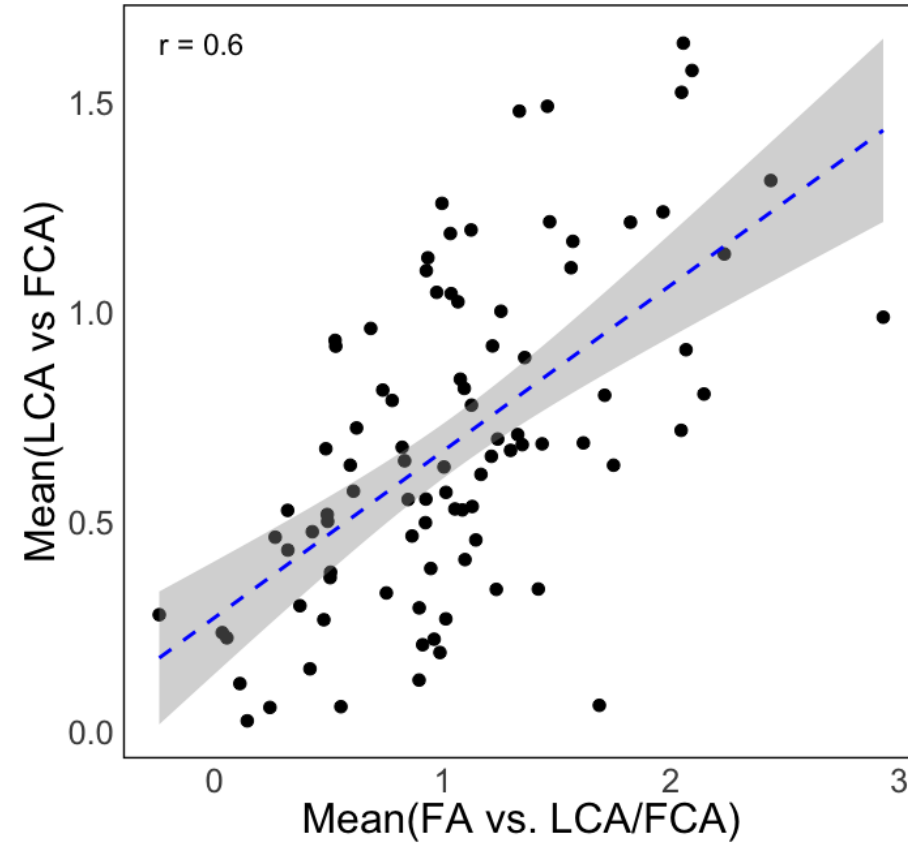
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Expt2: Correlation plots

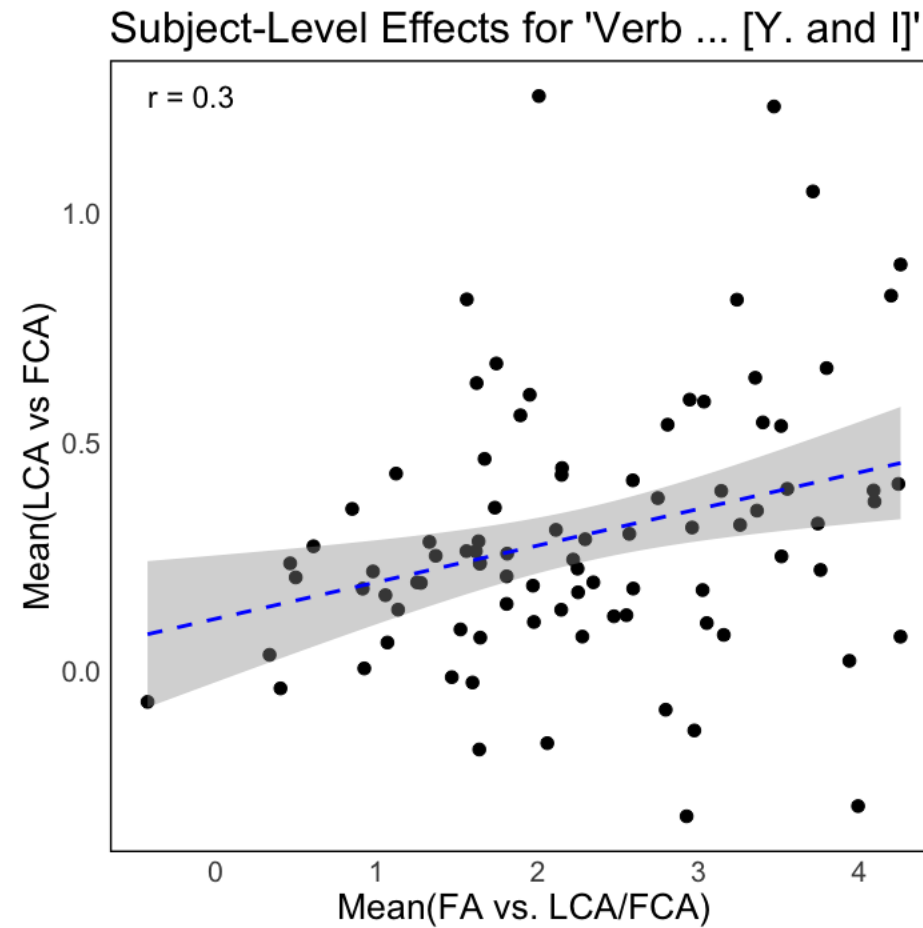
Subject-Level Effects for 'and'



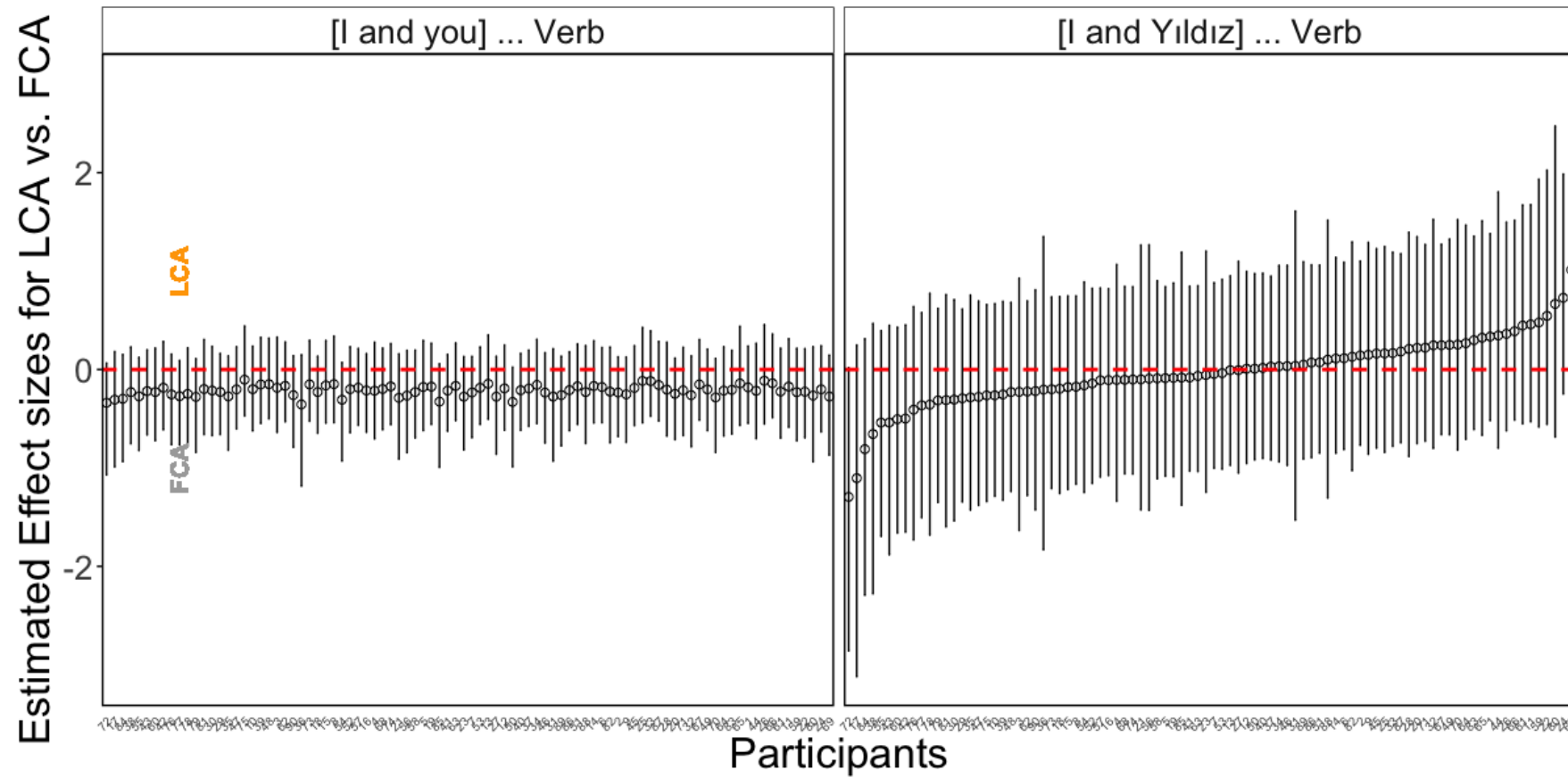
Subject-Level Effects for 'or'



Expt3: Correlation plots



Expt3: Speaker-level



Expt3: Correlation plots

